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Applied Information Extraction in Python

Today's goal • 3/3 items

Module 1
Information Extraction

Module 2
Named Entity Recognition (NER)

Named Entities and Named Entity Recognition

- What is Named Entity Recognition (NER)? Video • 5 min
- Knowledge Check: Named Entities and Named Entity Recognition Practice Assignment • Grade: 100%

Setting up NER as a Machine Learning Task

- BIO Encoding for Named Entity Labels Reading • 10 min
- **BILOU Encoding for Named Entity Labels** Reading • 10 min
- Machine Learning Fundamentals: How Machines Learn to Label Named Entities Reading • 10 min
- NER as a Sequence Classification Task Video • 8 min
- Knowledge Check: Setting up NER as a Machine Learning Task Practice Assignment • 10 min

Hidden Markov Models (HMMs)

- Markov Chain and Hidden Markov Models Reading • 10 min
- Fundamentals of Markov Chain Models Video • 7 min
- Hidden Markov Models (HMMs) Video • 11 min
- Training Hidden Markov Models: How HMMs Learn to Assign Labels Reading • 10 min
- The Math Behind HMMs: How Probabilities Power Sequence Labeling Reading • 10 min
- Conditional Random Fields (CRFs) Video • 6 min
- Knowledge Check: Hidden Markov Models (HMMs) Practice Assignment • 15 min

How to Extract Information Using CRFs in Python?

- Demo: CRF Model Training Video • 8 min
- Jupyter Notebook Practice on Training CRFs Lab • 1h

Module Wrap-Up & Assignment

- Assignment 2 Introduction: Implementing a CRF Model Video • 4 min
- Build an Information Extraction Pipeline for CRF Based Extraction Programming Assignment • 3h
- Module 2 Assignment Graded Assignment • 30 min

Module 3
Neural Network Models

Module 4
Transformers Transform Information Extraction

BILOU Encoding for Named Entity Labels

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BILOU

BILOU is another system that gives structure to your labels and allows models to distinguish the beginning, inside, and outside of multi-word entities.

What Is BILOU Encoding?

BILOU stands for:

- **B**eginning of a multi-word entity,
- **I**nside a multi-word entity
- **L**ast word of a multi-word entity
- **O**utside any entity
- **U**nit-length entity (a complete one-word entity)

Example 1:

“The University of Michigan is located in Ann Arbor.”

Suppose:

- “University of Michigan” is an organization (ORG), and
- “Ann Arbor” is a location (LOC).

Using **BILOU**:

- “The” → **O**
- “University” → **B**-ORG
- “of” → **I**-ORG
- “Michigan” → **L**-ORG
- “is” → **O**
- “located” → **O**
- “in” → **O**
- “Ann” → **B**-LOC
- “Arbor” → **L**-LOC

Example 2:

“Google is hiring.”

Here, “Google” is an organization (ORG).

Using **BILOU**:

- “Google” → **U**-ORG
- “is” → **O**
- "hiring" → **O**

Summing Up

BILOU encoding:

- Adds **clarity**: Google is a single-token entity, while University of Michigan and Ann Arbor have clearly marked starts and ends.
- Offers **greater precision**, especially during evaluation or model interpretation.
- Helps **avoid boundary-related errors;** e.g., incorrectly tagging “Michigan is” as a separate entity.

Deepen Your Understanding: External Guides & Tools

- **spaCy’s Named Entity Recognition Guide**
  - A hands-on intro to how modern NLP pipelines apply NER, with examples of labeling and pipelines.
  - https://spacy.io/usage/linguistic-features#named-entities
- **HuggingFace: Token Classification Task Overview**
  - Understand how transformer-based models use label encodings for NER tasks
  - https://huggingface.co/transformers/task_summary.html#token-classification
- **NLTK Chapter 7: Information Extraction**
  - Learn how classic Python NLP tools like NLTK handle chunking, tagging, and named entity recognition.
  - https://www.nltk.org/book/ch07.html

Reflection Prompt

Label the following sentence using BIO or BILOU: “I studied at the University of Michigan, Ann Arbor.”

Consider:

- How would you tag both the university name and the city?
- Does one scheme make it easier to understand entity boundaries? Why?

Assignment Preview

In Module 1 assignment, you’ll apply **BIO** and **BILOU** encoding schemes to label components of addresses such as street names, cities, and ZIP codes.

Understanding how **BIO** or **BILOU** structures entities will directly impact how well your CRF model learns to extract this information.

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